

MULTIMEDIA



UNIVERSITY

STUDENT ID NO

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TABLE NO AND VENUE:

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MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER OCTOBER/NOVEMBER 2025 - (TRIM ID 2530)

CMT1124 – MATHEMATICS II

(Foundation in Computing)

26 FEBRUARY 2026
9.00 a.m. – 11.00 a.m.
(2 Hours)

Question	Marks
SECTION A	/ 20
SECTION B	
1	/10
2	/10
3	/10
Total	50

INSTRUCTIONS TO STUDENT

1. This question paper consists of 15 pages (13 pages of questions and 2 pages of formula).
2. Attempt **ALL** questions, both in Section A and Section B.
3. Please write your answers for Section A in the box provided on Page 1.
4. Please write your answers for Section B in the space provided. All necessary working steps **MUST** be shown to obtain maximum marks

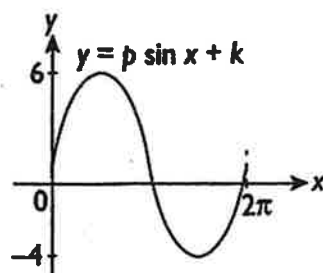
SECTION A [20 MARKS]

Please write your answer (A, B, C or D) in the box provided.

QUESTION	ANSWER
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Continued...

1. Given $\tan \theta = \frac{3}{4}$ and θ is in the third quadrant. Find the value of $\sin 2\theta$.
- A. $\frac{12}{25}$
B. $\frac{24}{25}$
C. $-\frac{12}{25}$
D. $-\frac{24}{25}$
2. The diagram shows the graph of $y = p \sin x + k$, where p and k are constant, for $0 \leq \theta \leq 2\pi$. The graph has a maximum value of 6 and a minimum value of -4 .



Determine the values of p and k .

	p	k
A.	5	1
B.	6	1
C.	5	2
D.	6	2

3. Solve the equation $4 \sin x = \sqrt{12} \tan x$ for $0^\circ \leq x \leq 360^\circ$.
- A. $0^\circ, 30^\circ, 180^\circ$
B. $0^\circ, 180^\circ, 360^\circ$
C. $0^\circ, 30^\circ, 180^\circ, 330^\circ$
D. $0^\circ, 30^\circ, 180^\circ, 330^\circ, 360^\circ$

Continued...

4. Simplify the following identity: $\csc 2A - \cot 2A$.

- A. $\cos A$
- B. $\sin A$
- C. $\tan A$
- D. $\cot A$

5. Find the exact value of $\sec \left[\csc^{-1} \left(-\frac{3}{2} \right) \right]$.

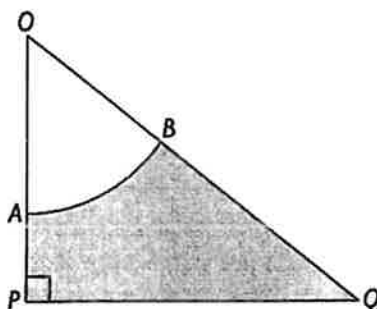
- A. $\frac{3\sqrt{5}}{5}$
- B. $-\frac{3\sqrt{5}}{5}$
- C. $\frac{5\sqrt{5}}{3}$
- D. $-\frac{5\sqrt{5}}{3}$

6. Given $w = \sqrt{3} + i$ and $z = 6 \left[\cos \left(\frac{5\pi}{3} \right) + i \sin \left(\frac{5\pi}{3} \right) \right]$. Find the polar form of $\frac{z}{w}$.

- A. $\frac{z}{w} = 3 \left[\cos \left(\frac{3\pi}{2} \right) + i \sin \left(\frac{3\pi}{2} \right) \right]$
- B. $\frac{z}{w} = 12 \left[\cos \left(\frac{3\pi}{2} \right) + i \sin \left(\frac{3\pi}{2} \right) \right]$
- C. $\frac{z}{w} = 3 \left[\cos \left(\frac{11\pi}{6} \right) + i \sin \left(\frac{11\pi}{6} \right) \right]$
- D. $\frac{z}{w} = 12 \left[\cos \left(\frac{11\pi}{6} \right) + i \sin \left(\frac{11\pi}{6} \right) \right]$

Continued...

7. The diagram below shows a sector AOB with a radius of 6 cm and center O inscribed in a right-angled triangle OPQ .

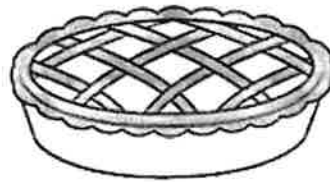


Given $\angle POQ = 0.9273$ rad and $OA = 2AP$. Find the perimeter of the shaded region.

- A. 12.564 cm
B. 15.564 cm
C. 25.564 cm
D. 29.564 cm
8. Based on the diagram in question 7, find the area of the shaded region.
- A. 16.691 cm^2
B. 34.691 cm^2
C. 37.309 cm^2
D. 54.310 cm^2
9. Given the complex number $z = \sqrt{3} - i$. Find the polar form of z^3 .
- A. $z = 2(\cos 270^\circ + i \sin 270^\circ)$
B. $z = 2(\cos 300^\circ + i \sin 300^\circ)$
C. $z = 8(\cos 270^\circ + i \sin 270^\circ)$
D. $z = 8(\cos 330^\circ + i \sin 330^\circ)$

Continued...

10. The diagram shows an apple pie. Amanda cuts the pie into 8 equal parts and gives 3 parts to her sister.



Given the surface area of the remaining apple pie is $\frac{1125}{8}\pi \text{ cm}^2$. Find the radius of the apple pie.

- A. 5
- B. 10
- C. 15
- D. 20

Continued...

SECTION B [30 MARKS]**QUESTION 1 [10 MARKS]**

a) Evaluate the following limits.

i) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$ [2 marks]

ii) $\lim_{x \rightarrow 0} \left[\frac{\sin(3x)}{x} + 2x^2 + 1 \right]$ [2 marks]

iii) $\lim_{x \rightarrow \infty} \frac{x^{\frac{1}{3}} - 5x + 3}{2x + x^{\frac{1}{3}} - 4}$ [2 marks]

Continued...

Extra space for Question 1**Continued...**

b) Given a piecewise function $f(x) = \begin{cases} x^2 + 1 & ; x < 2 \\ hx - 1 & ; x = 2 \\ \frac{3x - k}{2} & ; x > 2 \end{cases}$

i) Find the value of k if the limit exists at $x = 2$. [2 marks]

ii) Find $\lim_{x \rightarrow 2} f(x)$. [0.5 marks]

iii) Find the value of h if the function is continuous at $x = 2$. [1.5 marks]

Continued...

QUESTION 2 [10 MARKS]

a) Differentiate the following functions with respect to x and simplify your answer.

i) $y = (2x + 5)^3 (3x - 1)^4$ [3 marks]

ii) $y = \tan^3 4x$ [3 marks]

Continued...

- b) Find the absolute maximum and minimum values of the function

$$y = x^3 - 12x \text{ for } -3 \leq x \leq 5.$$

[4 marks]

Continued...

QUESTION 3 [10 MARKS]

a) Integrate the following integrals.

i) $\int_1^4 \left(8x^3 - 6\sqrt{x} + \frac{1}{x^3} \right) dx$ [2 marks]

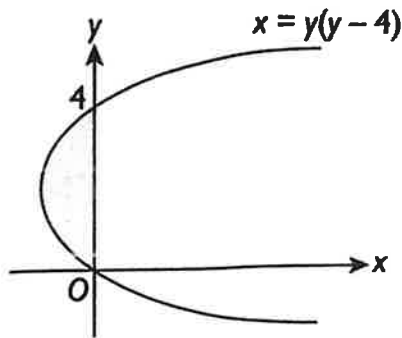
ii) $\int (x+1) \sin(x^2 + 2x) dx$ (Use integration by substitution method) [3 marks]

iii) $\int x^3 \ln(2x) dx$ (Use integration by parts method) [3 marks]

Continued...

Extra space for Question 3**Continued...**

- b) The diagram below shows the graph of function $x = y(y - 4)$.



Calculate the volume generated when the region bounded by the graph and the y -axis is rotated 360° about the y -axis.

[2 marks]

Blank area for the student's solution.

End of Question.

FORMULAE- CMT1124

$s = r\theta$	$A = \frac{1}{2}r^2\theta$
$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	$a^2 = b^2 + c^2 - 2bc \cos A$
$Area = \frac{1}{2}ab \sin C$	$Area = \sqrt{s(s-a)(s-b)(s-c)}$

Trigonometry

$\cos^2 A + \sin^2 A = 1$

$\sec^2 A = 1 + \tan^2 A$

$\csc^2 A = 1 + \cot^2 A$

$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

$\cos 2A = 2\cos^2 A - 1$ $\cos 2A = 1 - 2\sin^2 A$ $\cos 2A = \cos^2 A - \sin^2 A$ $\sin 2A = 2\sin A \cos A$ $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$	$\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}$ $\cos^2 \frac{A}{2} = \frac{1 + \cos A}{2}$ $\tan^2 \frac{A}{2} = \frac{1 - \cos A}{1 + \cos A}$	$\sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}$ $\cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}$ $\tan \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}} = \frac{1 - \cos A}{\sin A} = \frac{\sin A}{1 + \cos A}$
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$2\sin A \cos B = \sin(A + B) + \sin(A - B)$ $2\cos A \cos B = \cos(A + B) + \cos(A - B)$ $2\sin A \sin B = \cos(A - B) - \cos(A + B)$	$\sin A + \sin B = 2\sin \frac{A+B}{2} \cos \frac{A-B}{2}$ $\sin A - \sin B = 2\cos \frac{A+B}{2} \sin \frac{A-B}{2}$ $\cos A + \cos B = 2\cos \frac{A+B}{2} \cos \frac{A-B}{2}$ $\cos A - \cos B = -2\sin \frac{A+B}{2} \sin \frac{A-B}{2}$
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Differentiation Rules

$\frac{d}{dx}[x^n] = nx^{n-1}$	$\frac{d}{dx}[e^x] = e^x$	$\frac{d}{dx}[\ln x] = \frac{1}{x}; \quad x > 0$
$\frac{d}{dx}[\sin x] = \cos x$	$\frac{d}{dx}[\cos x] = -\sin x$	$\frac{d}{dx}[\tan x] = \sec^2 x$
$\frac{d}{dx}[\sec x] = \sec x \tan x$	$\frac{d}{dx}[\cot x] = -\csc^2 x$	$\frac{d}{dx}[\csc x] = -\csc x \cot x$

Product Rule:	$\frac{d}{dx}[f(x) \cdot g(x)] = f(x)g'(x) + f'(x)g(x)$
Quotient Rule:	$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$
Chain Rule:	$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$
Chain Rule combined with Power Rule	$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \cdot g'(x)$

Basic Integration Formulas

$\int cf(x)dx = c \int f(x)dx$	$\int k \cdot dx = kx + C$
$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$	$\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$
$\int e^x dx = e^x + C$	$\int \frac{1}{x} dx = \ln x + C$
$\int \sin x dx = -\cos x + C$	$\int \cos x dx = \sin x + C$
$\int \sec^2 x dx = \tan x + C$	$\int \sec x \tan x dx = \sec x + C$
Integration by-parts:	$\int u dv = uv - \int v du$
Area:	$\int_a^b (f(x) - g(x))dx$
Volume (disk):	$= \pi \int_a^b (f(x))^2 dx$
Volume (washer):	$\pi \int_a^b [(f(x))^2 - (g(x))^2] dx$

End of Page.